Cairo University
Faculty of Computers and Artificial Intelligence

Term Project

Department: Operations Research and Decision Support
Course Name: Systems Modeling and Simulation **Due Date: December 19th, 2025**
Course Code: DS331/DS241
Instructor: Assoc. Prof. Ayman Ghoneim

General Instructions to Students

- This term *project* is an assessment which contributes to the students' term grades.
- This is a *group* project for *four to five* students per group.
- The programming languages allowed to be used in the research project are Python, C++, or Java.
- Due date (tentative) is December 19th 2025 and Submission procedure and discussions will be announced later.
- For the submitted deliverables, see the end of the document.
- This document has *two* problems, and the group *must attempt both* problems.
- *All the project problems/tasks* must be attempted by *all the group members*.
- For each problem, it will be stated clearly what implementation is required and/or what should be included in the report.
- *Assessment* will be on the report documentation and code implementation submitted based on the following criteria:
  - The correctness of the algorithms employed and implementation.
  - The quality/comprehensiveness of your experiments & documentation.
  - The correctness of your analysis.
- ***Academic Integrity:*** You can only submit your own work. Any student/group suspected of plagiarism and/or using Artificial Intelligence tools (AI) generated content will be subject to the procedures set out in by the Faculty/University (including failing the course entirely) and punishments will extend to all the team members. Examples of behaviour that is not allowed are:
  - Copying all or part of someone else's work and submitting it as your own;
  - Using AI tools to generate (in whole or part) code, report or presentation.
  - Giving another student in the class a copy of your work; and
  - Copying parts from the internet, text books, etc without mentioning the associated references.

Problem I [Medical Clinic Multi-Channel Queue]

Patients arriving to a medical clinic belong to one of the following three categories, middle-age patients (younger than 60 years) with probability 0.7, elderly patients (older than 60 years) with probability 0.2, and emergency patients (critical case regardless of their age) with probability 0.1. The clinic has two doctors in two separate examination rooms. The first doctor examines the middle-age patients, while the second doctor examines the elderly patients. Since the elderly patients are less in numbers compared to the middle-age patients, the second doctor participates in examining middle-age patients if no elderly patients are waiting. The emergency patients have the highest priority, i.e., they will undergo examination once any of the two doctors is free. Some middle-age patient (with probability 0.3) may prefer to leave the clinic and visit later time if the waiting time was more than 30 minutes. Patients (regardless of their category) arrive to the medical clinic according to the inter-arrival time stated in Table 1. Middle-age and elderly patients have an examination time according to the examination-time distribution stated in Table 2, while emergency patients have an examination time according to the examination-time distribution stated in Table 3.

Table 1

Time between Arrivals (Minutes)	Probability
2	0.10
5	0.30
7	0.25
10	0.35

Table 2

Middle-Age & Elderly Patients Examination Time (Minutes)	Probability
5	0.20
7	0.30
10	0.50

Table 3

Emergency Patients Examination Time (Minutes)	Probability
7	0.20
10	0.50
15	0.30

The problem is to estimate the system measures of performance in terms of the following:

1- The average examination time of the three types of patients.
2- The average waiting time in the queues for each type of patients, and the average waiting time for all patients.
3- The maximum queue length for each doctor.
4- The probability that patients wait in each queue.
5- The portion of idle time of each doctor.

Moreover, the policy maker requires answers for the following questions:

6- Does the theoretical average examination time of the examination time distribution match with the experimental one for the three types of patients?
7- Does the theoretical average inter-arrival time of the inter-arrival time distribution match with the experimental one?
8- If the medical clinic is investigating the addition of one extra doctor in a new examination room, how will this affect the waiting time of the patients?

Assessments Marking Criteria
Problem I - The Medical Clinic

Simulation Project			
Report Components	Part 1 • Problem formulation & Objectives.	2	25
	Part 2 • System Components. • System analysis including cumulative distribution tables, calendar table (for 20 patients).	2 8	
	Part 3 • Experimental Design Parameters • Justification of experiment parameters values	2 2	
	Part 4 • Results Analysis: Using graphs & discussions stating the results for the 8 questions. • Conclusion	9	
Simulation Program	Coding Style (naming convention, comments, structure)	6	22
	GUI and Data Visualization (graphs)	4	
	Correct computation and results for • The average examination time of the three types of patients. • The average waiting time in the queues for each type of patients, and the average waiting time for all patients. • The maximum queue length for each doctor. • The probability that patients wait in each queue. • The portion of idle time of each doctor. • If the medical clinic is investigating the addition of one extra doctor in a new examination room, how will this affect the waiting time of the patients?	2 2 2 2 2 2	
Extra features in the simulator (for example: generic or extra statistics)			3
Total			**50**

Problem II [Electric Appliance Store]

An electric home appliance store sells two products A and B (e.g., refrigerators and washing machines), both products have the same size. The store has a show room that can accommodate a total of 10 units, and an inventory that can accommodate a total of 30 units. The daily demand for product A is given by the distribution shown in Table 1, while the daily demand for product B is given by the distribution shown in Table 2. When the how room runs out of a certain product, it sends a request for the inventory to receive a new 5 units from the product that ran out, and it receives instantaneously the 5 units or less according to the units' availability in the inventory. Each N days, the units in the inventory are revised to decide on an order (i.e., a review period). The store must decide on the quantity of product A (m_A) and the quantity of product B (m_B) to order carefully, since more units of one product means less available space for the other product. The lead time is the time from placing an order by the store until the order is received. The lead time in days is a random variable, as shown in Table 3. It is assumed that orders are placed at the close of business and are received for inventory at the beginning of business day as determined by the lead time. During the lead time, the store still receives demands for both products.

Table 1

Product A Demand (Units)	Probability
1	0.15
2	0.25
3	0.25
4	0.35

Table 2

Product B Demand (Units)	Probability
1	0.30
2	0.30
3	0.40

Table 3

Lead Time (Days)	Probability
1	0.4
2	0.35
3	0.25

Assuming that the inventory has already 30 units (15 of product A and 15 of product B), the store already has 5 units of product A and 5 units of product B, the policy maker wants to investigate the following assuming $N = 4$, $m_A = 6$ and $m_B = 6$.

1- The average ending units in the inventory and the showroom.

2- The number of days when a shortage condition occurs.

3- Do the theoretical average demands of product A and product B match the experimental ones?

4- Does the theoretical average lead time of the lead time distribution match the experimental one?

5- Is there a better value for the review period variable N to minimize the shortages in product A and product B?

6- Are there better values for m_A and m_B to minimize the shortages in product A and product B?

Assessments Marking Criteria
Problem II – Electric Appliance Store

<table>
<tr><th colspan="2">Simulation Project</th><th colspan="2"></th></tr>
<tr><td rowspan="4">Report Components</td><td>Part 1
• Problem formulation & Objectives.</td><td>2</td><td rowspan="4">25</td></tr>
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• System Components.
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8</td></tr>
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• Experimental Design Parameters
• Justification of experiment parameters values</td><td>2
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<tr><td>Correct computation and results for
• The average ending units in the inventory and the showroom.
• The number of days when a shortage condition occurs.
• Do the theoretical average demands of product A and product B match the experimental ones?
• Does the theoretical average lead time of the lead time distribution match the experimental one?
• Is there a better value for the review period variable N to minimize the shortages in product A and product B?
• Are there better values for m_A and m_B to minimize the shortages in product A and product B?</td><td>2
2
2
2
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2</td></tr>
<tr><td colspan="3">Extra features in the simulator (for example: generic or extra statistics)</td><td>3</td></tr>
<tr><td colspan="3">Total</td><td>50</td></tr>
</table>

Deliverables

One compressed file which must include a report documentation (Word or PDF file) and code implementation files, following the below details.

- Report documentation including:
  - Cover Sheet: Includes the CU and FCAI logos, course code, course name, problem title, group members (name and ID).
  - Table of Contents
  - Each requirement in the problem. Your report must be organized following the same organization of requirements and marking criteria stated here in the document.

- Code Implementation files, where each file is named after the part it corresponds to. For example, Problem I. The code file can be included in a folder (e.g., Problem I) if you are using several implementation code files.

Good Luck ☺